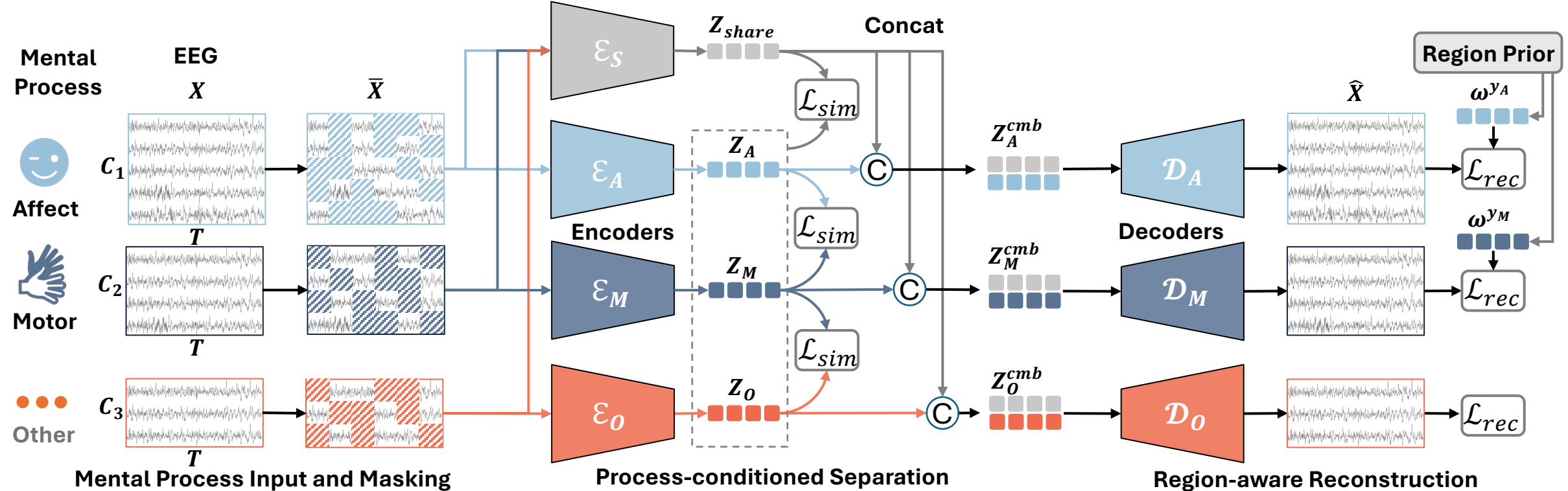

Mental Process
EEG
X
$\bar{X}$
Affect
Motor
Other
C_1
C_2
C_3
T
T
T
Mental Process Input and Masking
$\mathcal{E}_S$
$\mathcal{E}_A$
$\mathcal{E}_M$
$\mathcal{E}_O$
Encoders
Z_{share}
Z_A
Z_M
Z_O
$\mathcal{L}_{sim}$
$\mathcal{L}_{sim}$
$\mathcal{L}_{sim}$
Concat
C
C
C
Process-conditioned Separation
Z_A^{cmb}
Z_M^{cmb}
Z_O^{cmb}
$\mathcal{D}_A$
$\mathcal{D}_M$
$\mathcal{D}_O$
Decoders
$\hat{X}$
Region Prior
ω^{y_A}
ω^{y_M}
$\mathcal{L}_{rec}$
$\mathcal{L}_{rec}$
$\mathcal{L}_{rec}$
Region-aware Reconstruction